

06_GENERATION_ENGINE.md

SpeechStudio Generation Engine Specification

Version: 1.0

Status: Approved

1. Purpose

The Generation Engine is responsible for converting a validated prompt into synthesized speech using the local Higgs Audio V3 TTS model.

It is the only component allowed to communicate with the Transformers model.

2. Responsibilities

The Generation Engine shall

load the model

keep the model resident in memory

receive validated generation requests

invoke `generate_speech()`

receive generated audio

optionally append silence

save the audio

return generation metadata

The engine must never contain UI code.

3. Generation Pipeline

UI

↓

Prompt Builder

↓

Validation

↓

Generation Engine

↓

generate_speech()

↓

Audio Buffer

↓

Optional Silence Padding

↓

Save Output

↓

Audio Player

↓

History

4. Model Lifecycle

The model shall be loaded once during application startup.

The model shall remain in GPU memory until the application exits.

The model shall never be reloaded between generations.

The tokenizer shall be loaded once.

5. Supported Backend

Current backend

Transformers

Current model

multimodalart/higgs-audio-v3-tts-4b-transformers

Execution device

CUDA

Supported precision

FP16

BF16

6. Required Inputs

Prompt

Tokenizer

Loaded Model

Reference Audio (optional)

Reference Transcript (optional)

Generation Parameters

Output Path

7. Reference Voice

If voice cloning is enabled, the engine shall provide

reference_audio

reference_sample_rate

reference_text

to the model.

If voice cloning is disabled, these values shall not be supplied.

8. Generation Parameters

SpeechStudio shall expose the following parameters.

Temperature

Type

Float

Purpose

Sampling randomness

Recommended default

1.0

Supported UI range

0.10

to

2.00

Top P

Type

Float

Purpose

Nucleus sampling

Recommended default

0.95

Supported range

0.10

to

1.00

Top K

Type

Integer

Purpose

Limits candidate token selection

Recommended default

50

Supported UI range

1

to

500

Max New Tokens

Type

Integer

Purpose

Maximum generated token count

Recommended default

2048

Minimum

128

Maximum

4096

Seed

Type

Integer

Purpose

Optional deterministic generation

Default

Random

Append Silence

Type

Float

Purpose

Adds silence to the end of generated audio.

Recommended default

1.0 second

Range

0.0

to

5.0 seconds

Auto Play

Type

Boolean

Purpose

Automatically plays generated audio.

Normalize Output

Type

Boolean

Purpose

Normalize output volume before saving.

Output Format

Current

WAV

Future formats are outside the current project scope.

9. Audio Post Processing

The engine may optionally append silence after generation.

Purpose

Prevent abrupt audio ending.

Implementation

Append zero samples.

Duration configurable.

10. Output Naming

Default naming

yyyyMMdd_HHmmss.wav

Example

20260625_184512.wav

11. Output Directory

Default

outputs/

The engine shall create the directory if missing.

12. Error Handling

Possible failures

Model not loaded

CUDA unavailable

Reference audio missing

Reference transcript invalid

Generation failure

Output write failure

Every failure shall return a structured error.

No silent failures.

13. Generation Metadata

Each generation shall return

Generation Time

Output Duration

Realtime Factor

Output Sample Rate

Output File

Generation Parameters

Voice Profile

Timestamp

14. History Integration

After successful generation

Save audio

Save metadata

Register history entry

Return playback handle

15. Performance

The engine shall minimise unnecessary allocations.

Reference audio should be reused while the same voice profile remains active.

The tokenizer shall never be recreated during normal operation.

16. Memory Management

The engine shall

reuse the loaded model

reuse the tokenizer

release temporary tensors

avoid duplicate audio buffers where possible

17. CUDA Requirements

Preferred device

CUDA

If CUDA is unavailable

Generation shall be prevented unless CPU mode is explicitly enabled.

The application shall display the reason.

18. Logging

Each generation shall create a log entry containing

Start Time

End Time

Generation Duration

Generation Parameters

Voice Profile

Output File

Warnings

Errors

19. Benchmark Data

The engine shall expose

Generation Time

Output Length

Realtime Factor

GPU Name

VRAM Usage if available

These values are displayed in Benchmark mode.

20. Engine API

The UI communicates only with the Generation Engine.

The Generation Engine communicates only with the backend adapter.

The UI shall never directly call Transformers.

21. Configuration Ownership

The Generation Engine owns

generation parameters

sampling configuration

output writing

silence padding

metadata creation

The Prompt Builder owns prompt construction.

The Voice System owns reference voice management.

22. Threading

Generation shall execute on a worker thread.

The UI thread must remain responsive.

Audio playback shall not block generation.

23. Validation Requirements

Generation starts only if

Model loaded

Prompt valid

Voice valid

Reference audio valid if present

Output directory writable

Generation parameters valid

24. Supported Current Parameters

The engine currently supports

temperature

top_p

top_k

max_new_tokens

reference_audio

reference_sample_rate

reference_text

Future parameters shall only be added after confirmation from official Higgs documentation or verified implementation.

25. Success Criteria

The Generation Engine is successful if

the model is loaded once,

generation starts without Python scripting,

speech is produced,

metadata is stored,

history is updated,

and the UI remains responsive throughout the process.